CO1-AT-1

Short Answer Test

Computer Vision

ITA-0501

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Here, the main problem is very low spatial resolution which leads small objects to appear. To overcome this "Sampling" concept is applied which determines the no. of pixels used an image & makes increase of no. of pixels.

Increasing sampling rate using high-resolution camera will capture more details & improve image clarity even small object.

2. For a dark or noisy night image, use of histogram equalization image-processing technique is best as it improve contrast. Gaussian filter can be used to reduce noise & mak image more clear. These techniques make object more visible & improve quality of image.

3. Continuous intensity values can be converted into digital values using Quantization.

"Quantization" divide continuous intensity into fixed no. of discrete levels

Original Image → Sampling → Quantization → Digital Image

4. Uneven lighting can make a image of object to dark or too bright based on lighting. Here Histogram Equalization or contrast enhancement can be applied to improve the Product and defect visibility. It improves the intensity differences.

5. Here "Image Segmentation" can be used to divide image into regions such as roads vehicles, obstacles and the background. It helps the car to identify the objects and understand its surroundings
The information supports safe navigation and driving decisions

Edge Detection, such as canny edge detector, can be used to identify cracks in manufactured components. Cracks usually create sudden change in intensity & appear as edge in the image.

Detecting these edges helps the system identify defective products automatically

7. A median filter can be used to remove unwanted noise from medical image. It replaces noisy pixels using the median value of their neighboring pixels. This reduces noise while preserving important edges.

8. Object detection can identify vehicles in traffic image while object tracking follows their movement across video frames. This helps in counting vehicles and monitoring their movement. It is useful for detecting congestion & improving traffic management

9. The blur image is mainly caused by the movement of object or camera during the image acquisition, especially when shutter speed is slow. Here A faster shutter speed can reduce motion blur during image captur.

10. Image classification can be used to classify products good or defective. A model is trained using both the images good & defective product & learns their visual features. It can then automatically.

Product Image → Preprocessing → Feature Extraction

→ Image classification → Good / Defective product